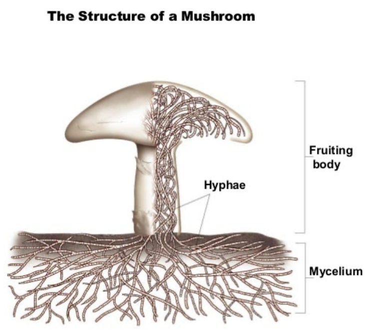
1. Growth of Mycelium block

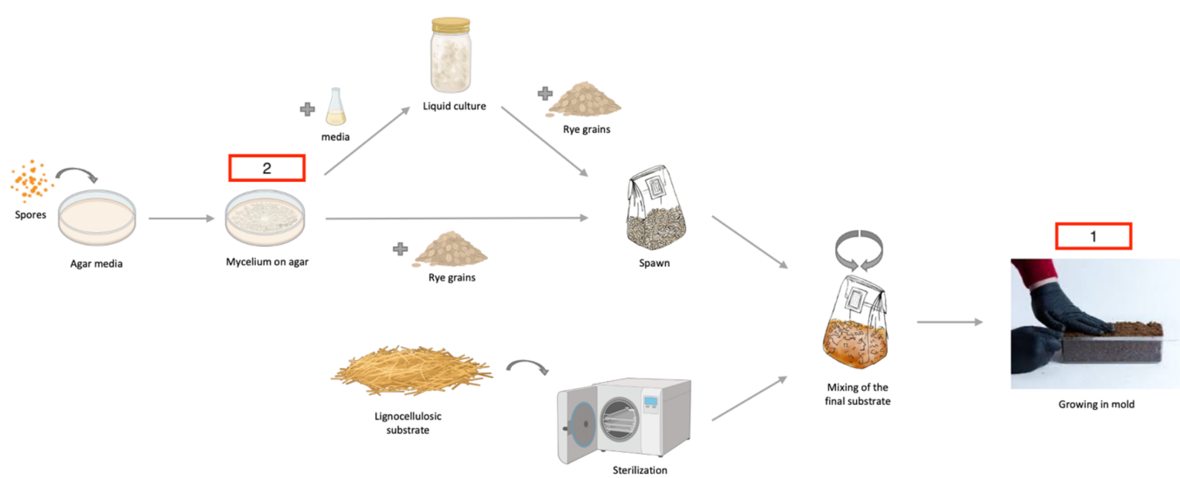
Basic knowledge about mycelium

What is mycelium?

- Root-like part of a fungus, consisting of thread-like strands called hyphae. Usually hidden underground or within a host (e.g. decaying wood).



What do we need to grow mycelium?



1. We first start from a living culture → inoculation (i.e. introducing fungus to its food)
    a. For structural composites, we buy commercial spawn (inoculated liquid culture with e.g. rye grain substrate) mixed with another substrate (food for fungus). → this document explains procedure
    b. For pure mycelium films, we propagate them on an agar plate from a mother culture (already inoculated mycelium on agar). → following document explains procedure.

Step by step

Materials

- Commercial spawn + substrate mix (ρ ~ 250 kg/m3)
- Mixing bucket
- Scale
- Flour
- Mold (not the bad fungus, the container)
- Ethanol
- Parafilm

Procedure

1. Turn on the light and the ventilation of the hood. UV light can be turned on for a few minutes for disinfection (should avoid contact with eyes and skin).

2. Sanitize the mixing bucket, the mold and the hood with ethanol.
3. Take the commercial mix and remove the outer white layer of the mycelium skin.
4. Measure the dimensions of the mold and calculate the approximate mass of mix needed.

    - 70 x 70 x 200mm moulds: approx. 250g per mould
    - 80 x 100 x 400mm moulds: approx. 800g per mould (approx. 600g with two slabs of wood)

5. Break the mix in the bucket as fine as possible.
6. Mix flour with the mix with a ratio of 30g flour/ kg of mix.
7. Put mixture into the mold and seal with parafilm (parafilm is sticky when it is stretched).
8. Place the mold into the growth chamber. 3-5 days is needed for growth (white brick-like appearance)
9. Make sure to check on it every 3 days, as a growth of a fruiting body can contaminate other experiments.
10. Wash and sanitize the equipment used with ethanol.